QUBO

$\min x^T Q x$ s.t. $x \in \{0,1\}^p$

Given the Example in [Binary LP.md]:

> $\min x_1^2 + x_2^2 + 5x_1x_2$ s.t. $x_1, x_2 \in \{0,1\}$
>
> $\equiv x_1 + x_2 + 5x_1x_2$ s.t. $x_1, x_2 \in \{0,1\}$

QUBO is a special case of linear programming because we can represent the multiplication by other variables.

> Note that Quantum Computers are REALLY good at solving these QUBO problems.

Lagrangian Relaxtion in QUBO

$L(\lambda) = \min c^T x - \lambda(f^T x - g)$

Boolean Condition	Linear Constraint	Quadratic Constraints
$x \vee y$	$x + y \geq 1$	$1 - x - y + xy = 0$
$x \vee y \vee z$	$x + y + z \geq 1$	$1 - x - y - z + xy + xz + yz - xyz = 0$
$x \to y$	$x \leq y$	$x - xy = 0$
$\neg(x \wedge y)$	$x + y \leq 1$	$xy = 0$
$x \wedge y$	$x + y = 2$	$1 - xy = 0$
$x \oplus y$	$x + y = 1$	$(1 - x - y)^2 =$

$\min x^T Q x$ s.t. $x \in \{0,1\}^q$

$= q_{11}x1^2 + q_{12}x_1x_2 + q_{21}x_1x_2 + ...$

$= q_{11}x1 + q_{12}x_1x_2 + q_{21}x_1x_2 + ...$

Vertex Cover

$\min \sum_{i \in V} x_i$

s.t. $x_i + x_j \geq 1, (\{i,j\} \in E)$ -> this constraint is a problem so apply lagrangian Relaxtion

> $x_i \in \{0,1\}, (i \in V)$

Lagrangian Relaxtion

$\min \sum x_i + \lambda \sum_{i,j \in E} (1 - x_i - x_j + x_i x_j)$

$= \sum_{i \in V} x_i (1 - \lambda deg(i)) + \lambda \sum_{i,j \in E} x_i x_j$